# Assignment #3

**#1:** $n$ - number of points, $K$ - max clusters.

For each $i = 1, \dots, n$ data point $(a_i, b_i) \in [0, 1000]^2$ (constants).

$p > 0$ - fixed penality per opened cluster.

Big-M constants: $M_x = M_y = 1000$ (valid since all coords in $[0, 1000]$).

Let $x_j \in [0, 1000]$ be the $x$-coordinate of center $j$, for $j = 1, \dots, K$.

Let $y_j \in [0, 1000]$ be the $y$-coordinate of center $j$, for $j = 1, \dots, K$.

Let $o_j \in \{0, 1\}$ indicate whether center $j$ is opened (1) or not (0).

Let $z_{ij} \in \{0, 1\}$ indicate whether point $i$ is assigned to center $j$ (1) or not (0),

for $i = 1, \dots, n$ and $j = 1, \dots, K$.

Let $u^x_{ij} \geq 0$ represent $|a_i - x_j|$ when $z_{ij} = 1$.

Let $u^y_{ij} \geq 0$ represent $|b_i - y_j|$ when $z_{ij} = 1$.

$$\min \; p \sum_{j=1}^{K} o_j + \sum_{i=1}^{n} \sum_{j=1}^{K} (u^x_{ij} + u^y_{ij}).$$

s.t. $\sum_{j=1}^{K} z_{ij} \;\; \forall i$ (each point picks exactly one center)

$z_{ij} \leq o_j \;\; \forall i, j$ (cannot assign to a closed center)

$$\left.\begin{aligned}
&u^x_{ij} \geq a_i - x_j - M_x(1 - z_{ij}) \quad \forall\, i = 1, \dots, n,\; j = 1, \dots, K \\
&u^x_{ij} \geq x_j - a_i - M_x(1 - z_{ij}) \quad \forall\, i = 1, \dots, n,\; j = 1, \dots, K \\
&u^y_{ij} \geq b_i - y_j - M_y(1 - z_{ij}) \quad \forall\, i = 1, \dots, n,\; j = 1, \dots, K \\
&u^y_{ij} \geq y_j - b_i - M_y(1 - z_{ij}) \quad \forall\, i = 1, \dots, n,\; j = 1, \dots, K \\
&u^x_{ij} \leq M_x z_{ij} \\
&u^y_{ij} \leq M_y z_{ij}
\end{aligned}\right\}$$

L1 Distance Linearization:

When $z_{ij} = 1$, these force $u^x_{ij} = |a_i - x_j|$ and $u^y_{ij} = |b_i - y_j|$.

When $z_{ij} = 0$, they allow $u^x_{ij} = u^y_{ij} = 0$, so unassigned pairs don't add cost.

**#2** a) **Sets and Data:**

Items $I = \{1, \dots, 20\}$ with weights $w_i$ (given).

People $P = \{F, G, H\}$ (Fred, Ginger, Heather).

Backpack capacity $B = 50$.

**Decision Variables:**

$y_{iF}, y_{iG}, y_{iH} \in \{0,1\}$: item $i$ assigned to Fred, Ginger, Heather resp.

$x_F, x_G, x_H \geq 0$: total carried weight of Fred, Ginger, Heather resp.

$H_F, H_G, H_H$: hapiness values for Fred, Ginger, Heather resp.

(Heather PWL) For segement $k = 1, \dots, 10$:

→ $z_k = \{0,1\}$: segement selection binary

→ $t_k \geq 0$: "length" travelled inside segment $k$ (in pounds).

**Linking weights and assignments:**

$$y_{iF} + y_{iG} + y_{iH} = 1 \quad \forall i \in I$$

$$x_F = \sum_{i \in I} w_i y_{iF}, \quad x_G = \sum_{i \in I} w_i y_{iG}, \quad x_H = \sum_{i \in I} w_i y_{iH} \quad \forall i \in I$$

$$x_F \leq B, \quad x_G \leq B, \quad x_H \leq B.$$

**Happiness (Fred + Ginger - linear):**

$$H_F = 100 - 1.7 x_F$$

$$H_G = 90 - 1.8 x_G$$

**Heather's hapiness:**

True function: $h(x) = 50 - 45 \tan^{-1}\left(\pi \cdot \frac{x - 25}{50}\right)$

we partition $[0,50]$ into 10 equal segements of length 5:

$[0,5], [5,10], \dots, [45,50]$.

Use secant lines on each segement $k$ with slope $\alpha_k$ and intercept $\beta_k$.

Let $L_k = [5(k-1), 5k]$. Enforce:

$$\sum_{k=1}^{10} z_k = 1, \quad 0 \leq t_k \leq 5 z_k \ \ \forall k, \quad x_H = \sum_{k=1}^{10} (x_{k-1} + t_k), \text{ with } x_{k-1} = 5(k-1) z_k.$$

ensures exactly one segement is active and $x_H$ lies in that segement. (Same in class).

**Segement Coefficients (explicit as we asked in piazza):**

For segement $k$ on $[5(k-1), 5k]$, heathers approx. hapiness is: $\hat{h}(x) = \alpha_k + \beta_k$, when $z_k = 1$.

[5]:

| | k | Interval | alpha_k | beta_k |
|---|---|---|---|---|
| 0 | 1 | [0,5] | -2.519127 | 70.864142 |
| 1 | 2 | [5,10] | -2.802291 | 72.279961 |
| 2 | 3 | [10,15] | -2.600638 | 70.263427 |
| 3 | 4 | [15,20] | -2.080390 | 62.459709 |
| 4 | 5 | [20,25] | -1.546538 | 51.782663 |
| 5 | 6 | [25,30] | -1.132696 | 41.436613 |
| 6 | 7 | [30,35] | -0.840751 | 32.678264 |
| 7 | 8 | [35,40] | -0.638504 | 25.599631 |
| 8 | 9 | [40,45] | -0.496778 | 19.930593 |
| 9 | 10 | [45,50] | -0.395294 | 15.363823 |

**Objective Function:**

$$\max \ H_F + H_G + H_H.$$

b)

**INPUT:**

```
\ Model A3_2_backpacking_PWL
Maximize
  H_F + H_G + H_H
Subject To
 assign_once[0]: y[0,F] + y[0,G] + y[0,H] = 1
 assign_once[1]: y[1,F] + y[1,G] + y[1,H] = 1
 assign_once[2]: y[2,F] + y[2,G] + y[2,H] = 1
 assign_once[3]: y[3,F] + y[3,G] + y[3,H] = 1
 assign_once[4]: y[4,F] + y[4,G] + y[4,H] = 1
 assign_once[5]: y[5,F] + y[5,G] + y[5,H] = 1
 assign_once[6]: y[6,F] + y[6,G] + y[6,H] = 1
 assign_once[7]: y[7,F] + y[7,G] + y[7,H] = 1
 assign_once[8]: y[8,F] + y[8,G] + y[8,H] = 1
 assign_once[9]: y[9,F] + y[9,G] + y[9,H] = 1
 assign_once[10]: y[10,F] + y[10,G] + y[10,H] = 1
 assign_once[11]: y[11,F] + y[11,G] + y[11,H] = 1
 assign_once[12]: y[12,F] + y[12,G] + y[12,H] = 1
 assign_once[13]: y[13,F] + y[13,G] + y[13,H] = 1
 assign_once[14]: y[14,F] + y[14,G] + y[14,H] = 1
 assign_once[15]: y[15,F] + y[15,G] + y[15,H] = 1
 assign_once[16]: y[16,F] + y[16,G] + y[16,H] = 1
 assign_once[17]: y[17,F] + y[17,G] + y[17,H] = 1
 assign_once[18]: y[18,F] + y[18,G] + y[18,H] = 1
 assign_once[19]: y[19,F] + y[19,G] + y[19,H] = 1
 xF_link: - 4 y[0,F] - y[1,F] - y[2,F] - 2 y[3,F] - 10 y[4,F] - 2 y[5,F]
   - 3 y[6,F] - 9 y[7,F] - 2 y[8,F] - 5 y[9,F] - 8 y[10,F] - 7 y[11,F]
   - 9 y[12,F] - 6 y[13,F] - 6 y[14,F] - 11 y[15,F] - 9 y[16,F] - 7 y[17,F]
   - 13 y[18,F] - 4 y[19,F] + x[F] = 0
 xG_link: - 4 y[0,G] - y[1,G] - y[2,G] - 2 y[3,G] - 10 y[4,G] - 2 y[5,G]
   - 3 y[6,G] - 9 y[7,G] - 2 y[8,G] - 5 y[9,G] - 8 y[10,G] - 7 y[11,G]
   - 9 y[12,G] - 6 y[13,G] - 6 y[14,G] - 11 y[15,G] - 9 y[16,G] - 7 y[17,G]
   - 13 y[18,G] - 4 y[19,G] + x[G] = 0
 xH_link: - 4 y[0,H] - y[1,H] - y[2,H] - 2 y[3,H] - 10 y[4,H] - 2 y[5,H]
   - 3 y[6,H] - 9 y[7,H] - 2 y[8,H] - 5 y[9,H] - 8 y[10,H] - 7 y[11,H]
   - 9 y[12,H] - 6 y[13,H] - 6 y[14,H] - 11 y[15,H] - 9 y[16,H] - 7 y[17,H]
   - 13 y[18,H] - 4 y[19,H] + x[H] = 0
 cap_F: x[F] <= 50
 cap_G: x[G] <= 50
 cap_H: x[H] <= 50
 H_F_def: 1.7 x[F] + H_F = 100
 H_G_def: 1.8 x[G] + H_G = 90
 one_segment: z[1] + z[2] + z[3] + z[4] + z[5] + z[6] + z[7] + z[8] + z[9]
   + z[10] = 1
 t_bounds[1]: - 5 z[1] + t[1] <= 0
 t_bounds[2]: - 5 z[2] + t[2] <= 0
 t_bounds[3]: - 5 z[3] + t[3] <= 0
 t_bounds[4]: - 5 z[4] + t[4] <= 0
 t_bounds[5]: - 5 z[5] + t[5] <= 0
 t_bounds[6]: - 5 z[6] + t[6] <= 0
 t_bounds[7]: - 5 z[7] + t[7] <= 0
 t_bounds[8]: - 5 z[8] + t[8] <= 0
 t_bounds[9]: - 5 z[9] + t[9] <= 0
 t_bounds[10]: - 5 z[10] + t[10] <= 0
 xH_PWL_position: x[H] - 5 z[2] - 10 z[3] - 15 z[4] - 20 z[5] - 25 z[6]
   - 30 z[7] - 35 z[8] - 40 z[9] - 45 z[10] - t[1] - t[2] - t[3] - t[4]
   - t[5] - t[6] - t[7] - t[8] - t[9] - t[10] = 0
 H_H_PWL_value: H_H - 70.864142 z[1] - 58.268506 z[2]
   - 44.25704699999999 z[3] - 31.253859 z[4] - 20.851903 z[5]
   - 13.119213 z[6] - 7.455734 z[7] - 3.251991 z[8] - 0.059473 z[9]
   + 2.424406999999999 z[10] + 2.519127 t[1] + 2.802291 t[2]
   + 2.600638 t[3] + 2.08039 t[4] + 1.546538 t[5] + 1.132696 t[6]
   + 0.840751 t[7] + 0.638504 t[8] + 0.496778 t[9] + 0.395294 t[10] = 0
Bounds
 H_F free
 H_G free
 H_H free
Binaries
 y[0,F] y[0,G] y[0,H] y[1,F] y[1,G] y[1,H] y[2,F] y[2,G] y[2,H] y[3,F]
 y[3,G] y[3,H] y[4,F] y[4,G] y[4,H] y[5,F] y[5,G] y[5,H] y[6,F] y[6,G]
 y[6,H] y[7,F] y[7,G] y[7,H] y[8,F] y[8,G] y[8,H] y[9,F] y[9,G] y[9,H]
 y[10,F] y[10,G] y[10,H] y[11,F] y[11,G] y[11,H] y[12,F] y[12,G] y[12,H]
 y[13,F] y[13,G] y[13,H] y[14,F] y[14,G] y[14,H] y[15,F] y[15,G] y[15,H]
 y[16,F] y[16,G] y[16,H] y[17,F] y[17,G] y[17,H] y[18,F] y[18,G] y[18,H]
 y[19,F] y[19,G] y[19,H] z[1] z[2] z[3] z[4] z[5] z[6] z[7] z[8] z[9] z[10]
End
```

**OUTPUT:** (See uploaded ipynb and lp file as well!)

```
Optimize a model with 41 rows, 86 columns and 201 nonzeros
Model fingerprint: 0x3b89f19b
Variable types: 16 continuous, 70 integer (70 binary)
Coefficient statistics:
  Matrix range     [6e-02, 7e+01]
  Objective range  [1e+00, 1e+00]
  Bounds range     [1e+00, 1e+00]
  RHS range        [1e+00, 1e+02]
Presolve removed 6 rows and 3 columns
Presolve time: 0.00s
Presolved: 35 rows, 83 columns, 173 nonzeros
Variable types: 10 continuous, 73 integer (70 binary)
Found heuristic solution: objective 61.9944170
Found heuristic solution: objective 62.1850050

Root relaxation: objective 6.639912e+01, 48 iterations, 0.00 seconds (0.00 work units)

    Nodes    |    Current Node    |     Objective Bounds      |     Work
 Expl Unexpl |  Obj  Depth IntInf | Incumbent    BestBd   Gap | It/Node Time

     0     0   66.39912    0    4   62.18500   66.39912  6.78%     -    0s
H    0     0                      64.7944170   66.39912  2.48%     -    0s
H    0     0                      66.0991230   66.39912  0.45%     -    0s
H    0     0                      66.1991230   66.39912  0.30%     -    0s
H    0     0                      66.2991230   66.39912  0.15%     -    0s
H    0     0                      66.3991230   66.39912  0.00%     -    0s

Cutting planes:
  Gomory: 2
  Cover: 1
  MIR: 1
  Zero half: 1
  RLT: 3

Explored 1 nodes (48 simplex iterations) in 0.02 seconds (0.00 work units)
Thread count was 8 (of 8 available processors)

Solution count 7: 66.3991 66.2991 66.1991 ... 61.9944

Optimal solution found (tolerance 1.00e-04)
Best objective 6.639912300000e+01, best bound 6.639912300000e+01, gap 0.0000%

=== Optimal Solution ===
Objective (total happiness): 66.399123

Fred:    items=[4, 6, 7, 9, 10, 14, 15, 18, 19, 20], weight=50.00, H_F=15.0000
Ginger:  items=[1, 11, 12], weight=19.00, H_G=55.8000
Heather: items=[2, 3, 5, 8, 13, 16, 17], weight=50.00, H_H(approx)=-4.4009

Heather true h(x_H) = -4.4009
Approx error        = 0.000012
Segment k           = 10 (interval [45, 50])
Local offset t_k    = 5.0000  (x_H = 45 + 5.0000 = 50.0000)
```

c) Define: Number of items per person $n_F = \sum_i y_{iF}$, $n_G = \sum_i y_{iG}$

Two weight thresholds: $\sum_i w_i y_{iG}$ and $\sum_i w_i y_{iH}$.

Let $M$ be large constant (eg 50 for weights, $M = 20$ for counts).

Include selector binaries $u_1, u_2, u_3 \in \{0,1\}$ with $u_1 + u_2 + u_3 \geq 1$.

ACTIVATE EACH CONDITION WITH BIG M:

**Fred ≤ 7 items and Ginger ≤ 5 items:**

$$n_F \leq 7 + M(1 - u_1)$$

$$n_G \leq 5 + M(1 - u_1).$$

**Ginger ≥ 10 lbs and Heather ≥ 15 lb:**

$$\sum_i w_i y_{iG} \geq 10 - M(1 - u_2)$$

$$\sum_i w_i y_{iH} \geq 15 - M(1 - u_2)$$

**Ginger's hapiness ≥ Fred's ≥ Heather's (approx):**

$$H_G - H_F \geq -M(1 - u_3)$$

$$H_F - H_H \geq -M(1 - u_3)$$

## #3

**a)** Let $x_1 \geq 0$ be the number of servings of milk (250mL)
Let $x_2 \geq 0$ be the number of servings of banana (100g)
Let $x_3 \geq 0$ be the number of servings of salmon (100g)
Let $x_4 \geq 0$ be the number of servings of broccoli (91g)

(P)

min $0.44x_1 + 0.20x_2 + 3.16x_3 + 0.72x_4$ (minimize total cost)

s.t. $8.5x_1 + 1.1x_2 + 27x_3 + 2.5x_4 \geq 67$ (protein daily requirements)

$2.4x_2 + 2.3x_4 \geq 30$ (fibre daily requirements)

$15x_1 + 2x_2 + 5x_3 + 64x_4 \geq 100$ (vitamin A daily requirements)

$45x_1 + 64x_3 \geq 100$ (vitamin daily requirements)

$30x_1 + 3x_4 \geq 100$ (calcium daily requirements)

**b)** Associate dual variables $y = (y_P, y_F, y_A, y_D, y_C)^T \geq \mathbf{0}$, for (Protein, fibre, vit A, vit D, calcium).

(D)

max $67y_P + 30y_F + 100y_A + 100y_D + 100y_C$

s.t. $8.5y_P + 15y_A + 45y_D + 30y_C \leq 0.44$ (milk)

$1.1y_P + 2.4y_F + 2y_A \leq 0.20$ (banana)

$27y_P + 5y_A + 64y_D \leq 3.16$ (salmon)

$2.5y_P + 2.3y_F + 64y_A + 3y_C \leq 0.72$ (brocolli)

**c)** $y_P(8.5x_1 + 1.1x_2 + 27x_3 + 2.5x_4 - 67) = 0$ (Protein)

$y_F(2.4x_2 + 2.3x_4 - 30) = 0$ (fibre)

$y_A(15x_1 + 2x_2 + 5x_3 + 64x_4 - 100) = 0$ (vitamin A)

$y_D(45x_1 + 64x_3 - 100) = 0$ (vitamin D)

$y_C(30x_1 + 3x_4 - 100) = 0$ (calcium)

$x_1(8.5y_P + 15y_A + 45y_D + 30y_C - 0.44) = 0$ (milk)

$x_2(1.1y_P + 2.4y_F + 2y_A - 0.20) = 0$ (banana)

$x_3(27y_P + 5y_A + 64y_D - 3.16) = 0$ (Salmon)

$x_4(2.5y_P + 2.3y_F + 64y_A + 3y_C - 0.72) = 0$ (broccoli)

**d)** Using Gurobi (.ipynb attached as a3q3.ipynb/lp as a3q3.lp), the optimal servings (Primal) are:

→ Milk $x_1 = 6.26$, Banana $x_2 = 12.50$, Salmon $x_3 = 0$, Brocolli $x_4 = 0$
→ Minimum cost ≈ \$5.26 (two decimal places)

Tight/Slack at this Solution:

→ Protein and Fibre are tight: $8.5x_1 + 1.1x_2 + 27x_3 + 2.5x_4 = 67$, $2.4x_2 + 2.3x_4 = 30$
→ Vitamin A, vitamin D, calcium are slack (> required).

Dual (shadow) prices from gurobi (reported as Pi- for ≥ rows, taking positive shadow prices as -Pi:

→ Protein $y_P = 0.05176$, Fibre $y_F = 0.05961$, Vitamin A $y_A = 0$, Vitamin D $y_D = 0$, Calcium $y_C = 0$

$\boxed{\text{Strong Duality Holds! } 67y_P + 30y_F \approx 5.26 = \text{Same as Primal cost!}}$

CS check: $x_1, x_2 > 0 \Rightarrow$ their dual constraints bind
$x_3 = x_4 = 0 \Rightarrow$ their dual constraints are slack
Protein/Fibre bind $\Rightarrow y_P, y_F > 0$; A/D/C slack $\Rightarrow y_A = y_D = y_C = 0$

**e)** Interpreting each dual variable as Jamie's maximum acceptable price per unit of that nutrient if buying supplements instead of food:

Dual Constraints (one per food):

For food $j$, $A_j^T y \leq c_j$.

Meaning: The total supplement value of that food's nutrients, priced at $y$, cannot exceed the food's shelf price $c_j$. Otherwise Jamie could arbitrage by buying the food and "selling" the nutrients back. Thus acceptable supplement prices must not make any food strictly dominated.

Strong Duality:

min (cost of foods to meet needs) = max (cost of supplements at prices $y$ ... acceptable to Jaime.)

At optimality, Jamie is indifferent between:
i) buying foods $x^*$, and
ii) buying exactly the required nutrients $y^*$.

The optimal $y^*$ supports the observed food market.

Complementary Slackness (both sides):

→ (Primal → Dual): If a food is purchased ($x_j > 0$), then its nutrient value equals its price: $A_j^T y = c_j$. Foods with $A_j^T y < c_j$ are NOT purchased.

→ (Dual → Primal): If a nutrient is oversupplied (strictly above requirement), its price must be zero (Jamie won't pay for extra). A nutrient with a binding requirement has positive price.

**f)** Let $v(b)$ be the optimal cost with requirement vector $b$. If the protein requirement is reduced by $\theta \geq 0$ to $b' = b - \theta e_{Prot}$, then:

$0 \leq v(b) - v(b') \leq \theta y^*_{Prot}$,

where $y^*_{Prot}$ is the optimal dual variable (shadow price) for the protein constraint at $b$. Moreover for sufficient small $\theta$ (so the optimal basis doesn't change), the reduction is exactly $\theta y^*_{Prot}$.

Proof:

→ Lower Bound (nonnegativity): The current optimal primal $x^*$ for $b$ is feasible for the relaxed problem $b'$ (constraints are easier), hence $v(b') \leq c^T x^* = v(b)$: Thus $v(b) - v(b') \geq 0$.

→ Upper Bound (via duality). For any dual-feasible $y$, weak duality gives $v(b') \geq b'^T y = b^T y - \theta y_{Prot}$.
Using the optimal dual $y^*$ at $b$:
$v(b') \geq b^T y^* - \theta y^*_{Prot} = v(b) - \theta y^*_{Prot}$,
So $v(b) - v(b') \leq \theta y^*_{Prot}$.

→ Tightness for small $\theta$: When $\theta$ is small enough that the optimal basis remains unchanged, LP sensitivity (envelope theorem) implies $v(b') = v(b) - \theta y^*_{Prot}$.

Hence, $0 \leq v(b) - v(b') \leq \theta y^*_{Prot}$ with equality for small $\theta$.

⇒ We found $y^*_{Prot} \approx 0.05176$ \$/g. Thus reducing protein by $\theta$ grams lowers cost by at most $0.05176\theta$ dollars. (0.05 rounded to 2 decimal places).